AUTOMORPHIC AND GEOMETRIC REFORMULATIONS OF THE ERDŐS–STRAUS EQUATION, AND A DIVISOR-CLASS BARRIER

ANONYMOUS

Abstract. For a prime p we collect exact reformulations of the Erdős–Straus equation $4/p = 1/x + 1/y + 1/z$ in automorphic and arithmetic-geometric language. Type I solutions are $\mathrm{SL}_2(\mathbb{Z})$ matrices with a prescribed entry, equivalently Heegner points of discriminant $-4d$ in a thin wedge, whose discriminant-$(-4dp^2)$ forms are T_p-Hecke translates. Type II solutions are Heegner points of discriminant $-4dp$ lying on an exact shell around a cuspidal geodesic. Further reformulations use orders in the quaternion algebra $B_{p,\infty}$, the algebra $(-d,-p)_{\mathbb{Q}}$, a CM torus, and an isotropic ternary form. The Bright–Loughran Brauer class is a genus character; this is folklore and goes back to Yamamoto. All these encodings reduce to one statement, which we prove: once two of the Elsholtz–Tao coordinates are fixed, the solutions are either a finite union of congruence classes of p, or the divisors of one explicit integer $N_\theta(p)$ in one residue class. The only representative-independent invariants of the automorphic objects that enter the count of solutions are congruence or genus data. We then record negative results, each with an exact scope. Fixed ternary embeddings reduce to Mordell coverings. Any detector must work at main term $p^{o(1)}$. Thickening a target is the same as averaging over n. The Type II lattice problem is everywhere locally soluble, and globally soluble up to one sign, even at odd squares. Under Schinzel's Hypothesis H, every argument that certifies the equation through a fixed finite set of slices fails infinitely often. We give the quantitative gaps to known equidistribution theorems, list the escape hatches that remain open, and report numerical evidence: $f_I(p) \geq 1$ and $f_{II}(p) \geq 1$ for all primes $p \leq 10^8$. The Erdős–Straus conjecture is *not* proved here. The negative statements concern the routes examined and known technology. They are not impossibility theorems for the conjecture.

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1. Introduction

The Erdős–Straus conjecture (ES) asserts that for every integer $n \geq 2$

$$\frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}, \qquad x, y, z \in \mathbb{N}, \tag{1.1}$$

is soluble. It suffices to treat primes $n = p$, and by Mordell's identities only primes in six classes modulo 840 (the squares) remain open. ES has been verified to very large bounds and holds for almost all n in a strong quantitative sense, but it is open for an individual large prime.

Why automorphic and geometric input. In earlier internal work of this campaign ([PO], and Theorems M and C of [PS]) we proved that *congruence-covering* arguments cannot settle ES pointwise. More precisely, under Schinzel's Hypothesis H, every correct bounded witness-producing procedure built from ring operations, floor division, sign tests and factorisations of the integers it meets fails for infinitely many primes $p \equiv 1 \pmod{24}$ [PS, Thm C]. This is the procedure-level form of the classical odd-square obstructions (Yamamoto, Mordell–Schinzel, Elsholtz–Tao [ET, Prop. 1.6], Bright–Loughran [BL, Cor. 1.4]). It leaves room for input that is *not* congruence data: spectral, automorphic or arithmetic-geometric information about a fixed prime p. This note collects what such input can see.

What this note contains.

(1) *Exact dictionaries* (§2, PROVED). Type I and Type II solutions are rewritten as Heegner points, SL_2 matrices, Hecke translates, shells around cuspidal geodesics, orders in definite quaternion algebras, divisors of an element of a CM field, and points on an isotropic ternary quadric. Several of these are folklore. We mark attributions explicitly; in particular, the identification of the Bright–Loughran Brauer class with a genus character is *not* new (Yamamoto 1965, via [BL, App. A]).
(2) *The divisor-class reduction* (§3, PROVED). Each encoding is a "slice" of the Elsholtz–Tao variety. Fixing two coordinates leaves either a congruence condition on p or the divisors of one integer $N_\theta(p)$ in one residue class (Lemma 3.1, Theorem 3.2).
(3) *Negative results with exact scope* (§4): ternary embeddings, the scale constraint, thickening, local solubility at odd squares, geometry, and Hypothesis-H-conditional emptiness of single targets.
(4) *Quantitative gaps* to known equidistribution theorems (§5, Assessment), *open problems* (§7), and *numerical evidence* (§8).

What this note does not contain. ES is not proved here, conditionally or unconditionally. We know of no standard conjecture (GRH, GLH, Ramanujan, Selberg's eigenvalue conjecture, twisted Linnik–Selberg) that is known to imply ES. The literature survey behind this note [LA] found no prior pointwise result for ES, conditional or not. Every negative statement below concerns the specific encodings and mechanisms analysed, or "known technology". None is a theorem that ES cannot be proved by automorphic methods. Words like "barrier" are used in that mechanism-scoped sense; where they summarise more than a proved statement, they are labelled Assessment.

Labels. Each statement carries one of the labels PROVED (a complete proof here, or in the cited internal file, relative to explicitly cited literature), CONDITIONAL (proved assuming a named hypothesis), CERTIFIED (an exact finite computation), EVIDENCE (numerics), or ASSESSMENT (a judgement that is not a theorem). Proofs of short items are given in full. For longer ones we give a sketch and a pointer to the internal source file, which also records its internal review. All internal files have been reviewed internally only; this note has not been externally refereed.

Notation. $p \geq 5$ is prime, $\mathcal{L} = \log p$, and $\chi_D = \left(\frac{D}{\cdot}\right)$ is the Kronecker symbol. For a negative discriminant D, $\chi_D(-1) = -1$. We write $[A, B, C]$ for $AX^2 + BXY + CY^2$, and $f_I(p)$, $f_{II}(p)$ for the numbers of Type I and Type II solutions in the normalisation of [ET, §2].

2. Exact dictionaries

2.1. **The Elsholtz–Tao normal form.** For an odd prime p, p divides exactly one or exactly two of x, y, z in any solution of (1.1) (it cannot divide all three, since $1/x' + 1/y' + 1/z' = 4$ has no positive solution). Following [ET, §2] and [RT2, §2.0], put

$$V_I(p) = \{(a,b,c,d,e) \in \mathbb{N}^5 : \ 4abd = pe + 1, \ a + b = ce\},$$
$$V_{II}(p) = \{(a,b,c,d,e) \in \mathbb{N}^5 : \ 4abd = p + e, \ a + b = ce\}.$$

An element of $V_I(p)$ gives the solution $(abdp, acd, bcd)$, and an element of $V_{II}(p)$ gives $(abd, acdp, bcdp)$: indeed $(c + p(a+b))/(abcdp) = c(1+pe)/(abcdp) = 4/p$ and $(cp + a + b)/(abcdp) = c(p+e)/(abcdp) = 4/p$. Conversely every solution arises in this way [ET, Props. 2.2, 2.6]. Hence

$$\mathrm{ES}(p) \iff V_I(p) \cup V_{II}(p) \neq \emptyset. \tag{2.1}$$

On $V_I(p)$ one has $p \nmid cd$ [O130, Prop. 2.1], [O131, §5]. The counts $f_I(p)$, $f_{II}(p)$ count primitive tuples ($\gcd(a,b,c) = 1$) [ET, (2.10)]. A non-primitive tuple (ga', gb', gc', d) is the primitive tuple $(a', b', c', g^2 d)$, so existence questions are insensitive to primitivity.

2.2. **Type I: SL_2 matrices, Heegner wedges and the Hecke operator T_p.** The following is [O123, Lemmas 2.1–2.2, 2.4]; parts (i)–(ii) of Lemma 2.1 are rearrangements of [ET, §2].

Lemma 2.1 (PROVED)**.** *Let $a, c, d, f \geq 1$ with $p \nmid cd$. The following are equivalent:* (i) *$p = 4acd - f$ and $f \mid 4a^2d + 1$;* (ii) *$f \mid ap + c$ and $p + f = 4acd$;* (iii) *$f \mid p^2 + 4c^2 d$ and $p + f = 4acd$. In particular, (ii) holds for some d iff $f \mid ap + c$ and $4ac \mid p + f$, and (iii) holds for some a iff $f \mid p^2 + 4c^2 d$ and $f \equiv -p \pmod{4cd}$. In that case $e = (4a^2d + 1)/f$, $b = (ap + c)/f$ and $g = (p^2 + 4c^2 d)/f = 4bcd - p$ are integers, $4abd - pe = 1$, $a + b = ce$, and*

$$\gamma = \begin{pmatrix} 4ad & p \\ e & b \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}).$$

Proof. (i)⇒(ii): $c(4a^2d+1) = a \cdot 4acd + c = a(p+f) + c$, so $f \mid ap + c$. (ii)⇒(i): with $4acd = p + f$ the same identity gives $f \mid c(4a^2d+1)$, and $\gcd(f, c) \mid \gcd(4acd - p, c) \mid p$; if $p \mid f$ and $p \mid c$ then $p \mid cd$, which is excluded. (i)⇔(iii): modulo f we have $4acd \equiv p$, hence $4c^2d(4a^2d + 1) \equiv p^2 + 4c^2 d$, and $\gcd(f, 4c^2 d) \mid p$, which is 1 unless $p \mid cd$. For the identities, put $e = (4a^2d + 1)/f$ and $b = ce - a$. Then

$bf = c(4a^2d+1) - a(4acd-p) = ap + c$ (so $b \geq 1$), $4abd = e \cdot 4acd - 4a^2d = e(p+f) - 4a^2d = pe+1$, and $(4acd-p)(4bcd-p) = 4c^2d(4abd-pe)+p^2 = p^2+4c^2d$, using $a+b = ce$; see [ET, (2.8)–(2.9)]. □

Write $Q_- = [f, -4ad, de]$ (discriminant $-4d$) and $Q' = [f, 4cd, dg]$ (discriminant $-4dp^2$).

Proposition 2.2 (Hecke form; PROVED)**.** *For every Type I tuple, $Q'(x,y) = Q_-(x+2bd\,y,\ p\,y)$, i.e. $Q' = Q_- \circ \left(\begin{smallmatrix} 1 & 2bd \\ 0 & p \end{smallmatrix}\right)$. Thus the discriminant-$(-4dp^2)$ form of a Type I solution is a T_p-Hecke translate of a level-d Heegner form of discriminant $-4d$, and the roots satisfy $z_{Q'} = p\, z_{Q_-} - 2bd$.*

Proof. The middle coefficient is $2f \cdot 2bd - 4adp = 4d(bf - ap) = 4cd$. The last is $4b^2d^2f - 8abd^2p + dep^2 = d\big(4bd(ap+c) - 8abdp + ep^2\big) = d\big(4bcd - p(4abd - pe)\big) = d(4bcd - p) = dg$. □

Proposition 2.3 (the wedge; PROVED, [O123, Prop. 3.1])**.** *Let $\Lambda_d = \{z = (2ad + i\sqrt{d})/f : a, f \geq 1,\ f \mid 4a^2d+1\}$, the roots of the forms $[f, -4ad, de]$. For $z = x+iy \in \Lambda_d$ and $c \geq 1$,*

$$\frac{2c\sqrt{d}\,x - \sqrt{d} - p\,y}{y} = 4acd - f - p \in \mathbb{Z}.$$

Hence (a,c,d,f) is a Type I tuple of p if and only if z lies in the wedge $R_c = \{x+iy : |2c\sqrt{d}\,x - \sqrt{d} - py| < y\}$ at the cusp $1/(2c)$. The edges of R_c are hypercycles of the geodesic $\{\mathrm{Re}\, z = 1/(2c)\}$, and R_c is a tube of hyperbolic width $\operatorname{arsinh} \frac{p+1}{2c\sqrt{d}} - \operatorname{arsinh} \frac{p-1}{2c\sqrt{d}} \approx 2/\sqrt{p^2+4c^2d}$.

Proof. For $z \in \Lambda_d$ we have $x/y = 2a\sqrt{d}$ and $\sqrt{d}/y = f$; substitute. An integer of absolute value < 1 is 0. The width is the difference of the distances of the two edges from the geodesic. □

The target is therefore at the *Planck scale*: its width is $\leq 2/p$, and $\asymp 1/p$ when $c\sqrt{d} \ll p$. The same holds in the T_p-translated picture, where the wedge sits at $-1/(2a)$ [O123, Prop. 3.2].

Lemma 2.4 (genus condition; PROVED; classical)**.** *If (a,c,d,f) is a Type I tuple of an odd n with $\gcd(n, 2cd) = 1$, then $\chi_{-4d}(-n) = 1$. For $n = p$ prime this says $\left(\frac{-d}{p}\right) = -1$: p is inert in $\mathbb{Q}(\sqrt{-d})$. For an odd square $n = m^2$ the condition fails for every d.*

Proof. Every prime $q \mid f$ divides $n^2 + 4c^2d$ and is prime to $2cd$, so $-4c^2d \equiv n^2$ is a non-zero square mod q, and $\chi_{-4d}(f) = 1$. As $f \equiv -n \pmod{4cd}$ and χ_{-4d} has period dividing $4d$, $\chi_{-4d}(-n) = \chi_{-4d}(f) = 1$. For $n = m^2$, $\chi_{-4d}(-m^2) = \chi_{-4d}(-1) = -1$. For $n = p$, $\chi_{-4d}(-p) = -\chi_{-4d}(p) = -\left(\frac{-d}{p}\right)$. □

This is the Type I half of [ET, Prop. 1.6], read at a fixed level. Any fixed-p main term at level d must therefore carry the factor $(1 + \chi_{-4d}(-p))/2$.

2.3. **Type II: determinant one at level dp, and shells.** The following is [O128, Lemma 1.1, Props. 2.1, 2.5, 7.1, Lemma 6.1].

Proposition 2.5 (PROVED)**.** *Let $c, d \geq 1$, $M = 4cd$ and $N_{c,d} = 4c^2dp + 1$.*

(a) *Type II tuples of p with these (c,d) correspond to factorisations $ff' - 4c^2(dp) = 1$ with $f \equiv f' \equiv -1 \pmod M$, via $f = 4acd-1$, $f' = 4bcd-1$. Equivalently, f runs over the divisors of $N_{c,d}$ with $f \equiv -1 \pmod M$ (the cofactor is then automatically $\equiv -1$).*

(b) *The map $(c,f) \mapsto Q = [f, 4cdp, dpf']$ sends these solutions injectively to Heegner forms of discriminant $-4dp$ at level dp. For fixed (p,d) the corresponding points of $\mathbb{H}$ are pairwise at hyperbolic distance with $\cosh \geq 3/2$.*

(c) *In the determinant-p coordinates $Q = [f, 4ad, de]$ (level one, discriminant $-4dp$), with $S_c = [0, 2c, 1]$ and the bilinear form $b(x,y) = q(x+y)-q(x)-q(y)$ of $q([A,B,C]) = B^2 - 4AC$, one has $q(Q) = -4dp$, $q(S_c) = 4c^2$ and $b(Q,S_c) = 4$. Thus the Type II solutions are embeddings of the binary lattice $4 \cdot [-dp, 1, c^2]$ (discriminant $4c^2dp+1$ before scaling) into the trace-zero lattice of signature $(2,1)$. Geometrically, a Type II point is a Heegner point of discriminant $-4dp$ at hyperbolic distance exactly $\operatorname{arsinh}\big(1/(2c\sqrt{dp})\big)$ from the cuspidal geodesic $(\infty, -1/(2c))$: it lies on a* shell.

(d) *(genus lemma) $(p/f) = (p/f') = -1$ (Jacobi symbols).*

Proof of (a), (c), (d). (a) $(4acd-1)(4bcd-1) = 4c^2d(4abd-e)+1 = 4c^2dp+1$ using $a+b = ce$ and $4abd = p+e$. Conversely, given $f \mid N_{c,d}$ with $f \equiv -1 \pmod M$, the cofactor f' satisfies $ff' \equiv 1$, hence $f' \equiv -1 \pmod M$; put $a = (f+1)/M$, $b = (f'+1)/M$; expanding $ff' = N_{c,d}$ gives $4abcd = a+b+cp$, so $c \mid a+b$ and $e = (a+b)/c = 4abd - p$. (c) With $e = (p+4a^2d)/f$ (an integer, [ET, (2.19)]), $q(Q) = 16a^2d^2 - 4def = -4dp$ and $b(Q,S_c) = 2 \cdot 4ad \cdot 2c - 4f = 4(4acd-f) = 4$. The distance formula is [O128, Prop. 2.5]. (d) $ef = p + 4a^2d$ gives $(2ad)^2 \equiv -dp \pmod f$ with $\gcd(dp, f) = 1$ ($f = 4acd-1$ is prime to d, and a prime $\ell \mid (p,f)$ would divide $2ad$, whence $f \equiv -1 \pmod \ell$), so $(-dp/f) = 1$. Since $f \equiv -1 \pmod{4d}$ and $\chi_{-4d}(-1) = -1$, $(-d/f) = \chi_{-4d}(f) = -1$, hence $(p/f) = -1$. The same argument applies to f'. □

Part (b) is [O128, Prop. 2.1, Lemma 2.2]; the separation is the lemma $\cosh \operatorname{dist} - 1 = \operatorname{disc}(Q-Q')/(8dp)$ of [TH, Lemma 2.2] applied at level dp. Part (d) recovers the Type II half of [ET, Prop. 1.6]: at an odd square $n = s^2$ one has $(s^2/f) = 1$. As for Type I, it is a level-wise reading of a classical fact; see Proposition 2.6.

2.4. **The Brauer class is a genus character (folklore).** Bright and Loughran [BL] study the surface $U_n : 4u_1u_2u_3 = n(u_1u_2 + u_1u_3 + u_2u_3)$ and its integral model. Over $\mathbb{Q}$ the group $\operatorname{Br}\tilde{U}_n / \operatorname{Br}\mathbb{Q}$ is $\mathbb{Z}/2$, generated by the transcendental quaternion algebra $\alpha = (-u_1/u_3, -u_2/u_3)$ [BL, Cor. 2.7]. They show that there is no Brauer–Manin obstruction to positive integral points for $n \geq 2$ [BL, Thms. 1.1, 1.8], that α obstructs strong approximation [BL, Thm. 1.8 (1.5), Thms. 1.2, 1.5] (though not every failure of strong approximation comes from α [BL, Thm. 1.9]), and that it recovers the odd-square obstruction [BL, Thm. 1.2, Cor. 1.4].

Proposition 2.6 (PROVED; folklore, Yamamoto [Ya] via [BL, App. A])**.** (a) *On the Type I locus at level d, the condition imposed by α is $\left(\frac{-ab}{p}\right) = \left(\frac{-d}{p}\right) = -1$, i.e. Lemma 2.4. For $\left(\frac{-d}{p}\right) = +1$ the positive adelic Brauer set of the level-d chart is empty.* (b) *On the Type II locus, α imposes $(f/p) = -(-1/p)$, which is equivalent to Proposition 2.5(d) and holds automatically. So the Brauer–Manin obstruction never removes a Type II level.*

Sketch. (a) $4abd = pe+1$ gives $(abd/p) = 1$, so $(-ab/p) = (-d/p)$. Lemma 2.4 gives $(-d/p) = -1$. For the local statement, $\mathrm{inv}_p\,\alpha = (-ab/p)$ on p-adic points of the chart, $\mathrm{inv}_\infty\,\alpha = -1$ on the positive part, and $\mathrm{inv}_\ell\,\alpha = +1$ for $\ell \neq p$ [BL, Lemmas 3.1, 3.5, 3.8]. (b) $e \equiv 4abd \pmod p$ gives $(-ab/p) = (-de/p)$, and $ef \equiv 4a^2d \pmod p$ gives $(e/p) = (d/p)(f/p)$. Since $f \equiv 3 \pmod 4$, reciprocity gives $(f/p) = (p/f)(-1/p)$. Details: [O126, Props. 5.1, 5.2]. □

Nothing here is new. Part (a) is the classical reason why the Mordell/Elsholtz–Tao families $n \equiv -f \pmod{4ad}$, $f \mid 4a^2d+1$, cover only non-residue classes. Part (b) is Yamamoto's lemma $(p/q) = -1$ for $q = 4abd-1$ in Mordell's notation, as derived from α in [BL, App. A]. (Yamamoto's paper was not accessed directly.)

2.5. **Quaternion orders.** For $d, m \geq 1$ let $\Lambda_{d,m} = \mathbb{Z}\langle u, v\rangle$ with $u^2 = -d$, $v^2 = -m$, $uv + vu = 1$. It is an order $\mathbb{Z} + \mathbb{Z}u + \mathbb{Z}v + \mathbb{Z}uv$ in the definite algebra $B_{d,m} \cong (-d, -d(4dm-1))_{\mathbb{Q}}$ (replace v by $2dv+u$, which anticommutes with u), of reduced discriminant $4dm-1$: the Gram matrix of the reduced trace on $(1, u, v, uv)$ has determinant $-(4dm-1)^2$ [O130, Lemma 1.2].

Proposition 2.7 (PROVED; [O130, Prop. 1.3])**.** *Let $(a,b,c,d,e) \in V_I(p)$ and $m = ab$. Then $B_{d,m} \cong B_{p,\infty}$, the quaternion algebra ramified exactly at p and ∞, and $\Lambda_{d,m}$ is an order of reduced discriminant $4abd-1 = pe$ in it. Consequently there is a supersingular elliptic curve over $\overline{\mathbb{F}}_p$ with endomorphisms u, v satisfying $u^2 = -d$, $v^2 = -ab$, $uv + vu = 1$.*

Sketch. Compute the Hilbert symbols of $(-d, -dpe)$. At $\ell \nmid 2dpe$ they are trivial. At p: $p \nmid e$ (if $p \mid e$ then $-4a^2d \equiv 1 \pmod p$, contradicting $(-d/p) = -1$), so the symbol is $(-d/p) = -1$ by Lemma 2.4. At odd $\ell \mid e$: $-d \equiv (2a)^{-2}$ is a square mod ℓ ($e \mid 4a^2d+1$), so the symbol is 1. At odd $\ell \mid d$: $pe \equiv -1 \pmod \ell$ gives 1. At ∞ it is -1, and the product formula gives 2. The last sentence is Deuring's correspondence. □

This was checked on all Type I tuples with $p < 3000$ (EVIDENCE, [O130]). In the language of Gross–Zagier [GZ], the identity $(16abd-4)/4 = pe$ has the shape $(D_1D_2 - x^2)/4$ with $D_1 = -4d$, $D_2 = -4ab$, $x = 2$, so the CM points of these discriminants have a common supersingular reduction. This is a *formal analogy only*: the theorem of [GZ] assumes coprime fundamental discriminants, and $-4d, -4ab$ are never coprime. No Gross–Zagier formula is used.

Lemma 2.8 (PROVED; [O130, Lemma 3.1])**.** *In $B = (-d, -p)_{\mathbb{Q}}$ with $i^2 = -d$, $j^2 = -p$, $k = ij = -ji$, Type II at level d holds if and only if, for some $a, b, c, e \geq 1$,*

$$(2a\,i + j)(2b\,i - j) = -e(1 + 2c\,k).$$

Proof. The left side is $-4abd - 2(a+b)k + p = -(4abd - p) - 2(a+b)k$; compare coefficients with $4abd - p = e$, $a + b = ce$. □

Taking reduced norms gives $\mathrm{N}(2ai + j) = 4a^2d + p = ef$ and $\mathrm{N}(1 + 2ck) = 1 + 4c^2dp = ff'$, the integers of the ternary form below and of the shells of Proposition 2.5. The novelty of the quaternion dictionary is *unverified*: the survey [LA] found no prior occurrence, but we do not claim it is new.

2.6. **The CM torus.** Let $p \nmid 2cd$, $K = \mathbb{Q}(\sqrt{-d})$, $\alpha = p+2c\sqrt{-d}$, so $\mathrm{N}\,\alpha = p^2+4c^2d$.

Proposition 2.9 (PROVED; [O131, Lemma 5.1, Thm. 5.2])**.** (i) $\mathfrak{a} \mapsto \mathrm{N}\,\mathfrak{a}$ *is a bijection from the ideal divisors of* (α) *in* $\mathbb{Z}[\sqrt{-d}]$ *to the positive divisors of* $\mathrm{N}\,\alpha$. (ii) *The number* $T_{c,d}(p)$ *of ordered Type I tuples with these* (c,d) *equals* $\#\{\mathfrak{a} \mid (\alpha) : \mathrm{N}\,\mathfrak{a} \equiv -p \pmod{4cd}\}$*, and*

$$T_{c,d}(p) = \frac{1}{\varphi(4cd)} \sum_{\chi \bmod 4cd} \bar{\chi}(-p)\,\sigma_\chi(p^2+4c^2d), \qquad \sigma_\chi(N) = \sum_{f \mid N} \chi(f).$$

(iii) *Only Hecke characters of the form* $\chi \circ \mathrm{N}$ *occur. In particular every class-group character of* K *that is not a genus character has coefficient* 0*, and pairing* $\chi \leftrightarrow \chi\chi_{-4d}$ *produces the factor* $1+\chi_{-4d}(-p)$ *of Lemma 2.4.*

Sketch. α is primitive since $\gcd(p,2c)=1$. Every prime $\ell \mid \mathrm{N}\,\alpha$ is odd and prime to d, so it is unramified and not inert, and it splits. Exactly one of the two primes above ℓ divides α, to the full power. This gives (i). Then (ii) is Lemma 2.1(iii) plus orthogonality on $(\mathbb{Z}/4cd)^\times$, and (iii) is genus theory. □

Thus the dihedral (Frobenius) information carried by the class of (f,α) in the ring class field is invisible in the count.

2.7. **The isotropic ternary form Q_d.** Let $Q_d(e,f,a) = ef - 4da^2$.

Proposition 2.10 (PROVED; [O124, Prop. 6.2], [RT1, §6])**.** (i) *Type II at level* d *with parameter* a *holds if and only if* $Q_d(e,f,a) = p$ *with* $4ad \mid f+1$ *and* $(f+1)e/(4ad) > a$*, i.e.* $p+4a^2d$ *has a divisor* $f \equiv -1 \pmod{4ad}$*. (The cofactor condition is automatic: with* $c=(f+1)/4ad$ *one has* $ce-a = (fp+p+4a^2d)/(4adf) > 0$*.)* (ii) Q_d *represents every integer primitively over* $\mathbb{Z}$ *(*$(e,f,a)=(n,1,0)$*), so local conditions, genus and spinor genus impose nothing; all the arithmetic sits in the side condition* $4ad \mid f+1$. (iii) *The side condition is not a finite union of lattice cosets: its slice at* $a=a_0$ *has* f*-period* $4a_0d$*, which is unbounded.* (iv) *On such points the cofactor* $f = 4acd-1$ *is a Type II* multiplier witness modulus*; in particular* $f \geq W(p)$*, the least such modulus.*

Among all the parametrisations of [ET, §2], Q_d is the only rank-3 quadratic piece (CERTIFIED by a systematic search, [O124, Comp. 6.1], reproduced independently in its review).

3. The divisor-class reduction

This section is [RT2, §2]. A *slice* is a choice θ of two of the coordinates $\{a,b,c,d,e\}$ together with prescribed values for them; $V_\theta(p)$ denotes the elements of $V_I(p)$ (or $V_{II}(p)$) with these values. By the symmetry $a \leftrightarrow b$ the coordinate pairs are $\{a,b\},\{a,c\},\{a,d\},\{a,e\},\{c,d\},\{c,e\},\{d,e\}$.

Lemma 3.1 (slice table; PROVED, elementary)**.** *For each type and each slice* θ *exactly one of the following holds.*

- (C) Congruence slice: $V_\theta(p) \neq \emptyset$ *if and only if* p *lies in an explicit finite union of residue classes modulo a modulus depending only on* θ*.*
- (D) Divisor slice: *there are an integer* $N_\theta(p)$ *(a polynomial in* p *of degree* ≤ 2*, possibly divided by a constant depending on* θ*), a modulus* q_θ *and a residue set* $R_\theta(p) \subset \mathbb{Z}/q_\theta$ *such that* $V_\theta(p)$ *is in bijection with* $\{f \geq 1 :\ f \mid$

$N_\theta(p),\ f \bmod q_\theta \in R_\theta(p)\}$. *The set* $R_\theta(p)$ *depends only on* θ, *on* $p \bmod q_\theta$ *and, in the* $\{d,e\}$ *rows, on* $N_\theta(p) \bmod q_\theta$.

The rows are as follows (Type I rows containing c *or* d *are for* $p \nmid cd$; *otherwise the fibre is empty).*

type	θ	*kind*	$N_\theta(p)$	*divisor* f	*class condition*
I	$\{c,d\}$	*D*	p^2+4c^2d	$4acd-p$	$f \equiv -p\ (4cd)$
I	$\{a,c\}$	*D*	$ap+c$	$4acd-p$	$f \equiv -p\ (4ac)$
I	$\{a,e\}$	*D*	$(pe+1)/4a$	b	$f \equiv -a\ (e)$
I	$\{d,e\}$	*D*	$m=(pe+1)/4d$	a	$f^2 \equiv -m\ (e),\ (f,e)=1$
I	$\{a,d\}$	*C*	$4a^2d+1$	$4acd-p$	$p \equiv -f\ (4ad)$
I	$\{a,b\},\{c,e\}$	*C*	—	—	p *mod* $4ab$, *resp.* $4a(ce-a)$
II	$\{c,d\}$	*D*	$4c^2dp+1$	$4acd-1$	$f \equiv -1\ (4cd)$
II	$\{a,c\}$	*D*	$cp+a$	$4acd-1$	$f \equiv -1\ (4ac)$
II	$\{a,d\}$	*D*	$p+4a^2d$	$4acd-1$	$f \equiv -1\ (4ad)$
II	$\{a,e\}$	*D*	$(p+e)/4a$	b	$f \equiv -a\ (e)$
II	$\{d,e\}$	*D*	$m=(p+e)/4d$	a	$f^2 \equiv -m\ (e),\ (f,e)=1$
II	$\{a,b\},\{c,e\}$	*C*	—	—	p *mod* $4ab$, *resp.* $4a(ce-a)$

In the $\{a,e\}$ *and* $\{d,e\}$ *rows,* $N_\theta(p)$ *is required to be an integer (otherwise the fibre is empty).*

Proof. Each D row is an identity plus a converse. $I\{c,d\}$ is Lemma 2.1(iii). For the converse, the cofactor $f'=N/f$ satisfies $f' \equiv p^2(-p)^{-1} \equiv -p \pmod{4cd}$; put $a=(f+p)/4cd$, $b=(f'+p)/4cd$; expanding $ff'=N$ gives $p(a+b)=c(4abd-1)$, so $c \mid a+b$ and $e=(a+b)/c=(4abd-1)/p$. $II\{c,d\}$ is Proposition 2.5(a). $I\{a,c\}$ is Lemma 2.1(ii). $II\{a,c\}$, $II\{a,d\}$: with $f=4acd-1$ one has $ef=4a^2d+p$, and $c(4a^2d+p)=a(f+1)+cp$, so $f \mid cp+a$ (as $\gcd(f,c)=1$). Conversely put $d=(f+1)/4ac$, resp. $c=(f+1)/4ad$, then $e=(p+4a^2d)/f$ and $b=ce-a$. $\{a,e\}$, $\{d,e\}$: with $M=pe+1$ (Type I) or $M=p+e$ (Type II) the defining equations read $bd=M/4a$, resp. $ab=M/4d$, together with $e \mid a+b$. For $\gcd(a,e)=1$, $e \mid a+m/a$ is equivalent to $e \mid a^2+m$, and every element has $\gcd(a,e)=1$: for Type I since $a \mid pe+1$; for Type II since $\gcd(a,e) \mid p$ and $p \nmid e$ (if $p \mid e$ then $4abd=p+e \le 2e \le 2(a+b)$ forces $a=b=d=1$ and then $4=p+e \ge 2p$, impossible). *C rows*: for $I\{a,d\}$ the fibre is $\{f \mid 4a^2d+1 : f \equiv -p\ (4ad)\}$, a condition on p mod $4ad$; for $\{a,b\}$, e runs over the divisors of $a+b$ and the condition is $4ab \mid pe+1$ (resp. $p+e$); for $\{c,e\}$, a runs over $[1,ce-1]$ with condition $4a(ce-a) \mid pe+1$ (resp. $p+e$). □

All bijections were verified mechanically against a brute-force enumeration of $V_I(p) \cup V_{II}(p)$ for every prime $5 \le p < 200$ (CERTIFIED, [RT2, Lemma 2.1]; script `scripts/rt2_slices.py`).

Theorem 3.2 (divisor-class reduction)**.** *Consider the encodings of §2: E1 the Type I* SL_2*/Heegner wedge (Proposition 2.3,* $\theta=\{c,d\}$*; its* T_p*-translated form, per tube,* $\theta=\{a,d\}$*); E2 the Type II shells (Proposition 2.5,* $\theta=\{c,d\}$*); E3 the quaternion order of Proposition 2.7 (*$\theta=\{d,e\}$*); E4 the* $(-d,-p)$ *factorisation of Lemma 2.8 (*$\theta=\{d,e\}$*; the norms of* $2ai+j$ *give* $\theta=\{a,d\}$*); E5 the CM torus (Proposition 2.9,* $\theta=\{c,d\}$*); E6 the ternary form* Q_d *(Proposition 2.10,* $\theta=\{a,d\}$*).*

(i) (PROVED) *The E-solutions with parameter* θ *are in bijection with* $V_\theta(p)$. *Hence they correspond to the divisors of the single integer* $N_\theta(p)$ *lying in* $R_\theta(p)$, *or (C rows) to a congruence condition on* p.

(ii) (PROVED, RELATIVE TO THE CITED INTERNAL STATEMENTS) *The only representative-independent invariants of these objects that enter the count of ES points are congruence or genus data, in the following precise senses: for E5 only genus characters have non-zero coefficient (Proposition 2.9). For E3 the collision datum is a function of θ, of $p \bmod 4d$, of $(-d/p)$ and of $(-d/\ell)$ for $\ell \mid e$, and it holds automatically on every admissible θ* [O130, Prop. 2.2]. *For E1, E2 and E6 the property of being an ES point belongs to the representative, not to the CM point or Γ-orbit: each class has infinitely many representatives of the required shape and only finitely many good ones (*[O126, Prop. 4.1]*;* [O128, Prop. 8.2]*; the stabiliser of S_c in $\mathrm{SL}_2(\mathbb{Z})$ is $\pm I$). For E4 the ES elements lie on the line $\{2xi \pm j\}$, on which the reduced norm is the one-variable polynomial $4dx^2 + p$* [O130, Prop. 3.2].

(iii) (CONDITIONAL ON HYPOTHESIS H) *Let Θ_0 be a finite set of slice parameters, of any types and rows, fixed independently of p. Any argument that certifies* $\mathrm{ES}(p)$ *by exhibiting an E-solution with parameter in Θ_0 fails for infinitely many primes $p \equiv 1 \pmod{24}$. This holds whatever auxiliary automorphic, spectral or geometric information the argument uses.*

Proof. (i) Each source file proves the exact lift from its object to $V_\theta(p)$ (Propositions 2.3, 2.5, 2.7, 2.9, 2.10 and Lemma 2.8); then apply Lemma 3.1. (ii) is a collection of the cited statements. (iii) The argument can only succeed at primes with $\bigcup_{\theta\in\Theta_0} V_\theta(p) \neq \emptyset$. By Lemma 3.1 this predicate is decided, together with a witness, by the bounded and correct procedure "for $\theta \in \Theta_0$: test $p \bmod q_\theta$; factor $N_\theta(p)$ (with floor division for the $\{\cdot, e\}$ rows); test each divisor against $R_\theta(p)$". Listing all divisors of an integer and looping over a list are primitive instructions of a bounded program in the sense of [PS, §1.2], so this procedure is bounded although the number of divisors grows with p. By [PS, Thm. C], under Hypothesis H every such procedure fails for infinitely many $p \equiv 1 \pmod{24}$. □

Remark 3.3 (scope)**.** Part (iii) says that whatever an automorphic input "knows", if it certifies ES through a fixed finite set of fibres it is no stronger than factoring finitely many fixed polynomials in p, and under H that fails. It does *not* say that automorphic methods cannot aggregate over a parameter set $\Theta(p)$ that grows with p; Kuznetsov's formula and Heegner equidistribution aggregate over levels and discriminants in exactly this way. Aggregation over θ is legitimate, since θ is a coordinate of the solution and not fixed by the problem. By Lemma 3.1 it amounts to controlling, at one prime p, the divisor events of a growing family $\{N_\theta(p)\}_{\theta\in\Theta(p)}$ jointly. The quantitative obstacles to doing so are in §§4–5, and the open routes in §7. Unconditional quantitative versions of (iii) exist for specific growing boxes. For the Type II box $\{acd \le T/4\}$ one uses $f = 4acd - 1 \ge W(p)$ together with the Ω-result $W(p) > \exp\big(c(\log p)^{1/4}(\log\log p)^{-1/4}\big)$ for infinitely many p [SE]. For Type I heights there is $ck_{\min}(p) \gg \log p \cdot \log\log\log p$ infinitely often (internal file `POINTWISE_OMEGA.md`, Thm. 8.5, modulo Lau–Wu). Both are cited as summarised in [RT2, §2.2].

4. Negative results with exact scope

Each result in this section concerns a precisely defined mechanism. None of them says that ES is false or unprovable.

4.1. **Ternary and higher-rank embeddings.** Fix $k \geq 3$, a non-degenerate integral quadratic form Q in k variables, and integers $\kappa \geq 1$, η, all independent of p.

Definition 4.1 ([O124, Def. 1.1])**.** An *embedding of complexity* (J, D) is a family of pieces $(S_j, \Phi_j)_{j \leq J}$, where $S_j \subseteq \mathbb{Z}^k$ is an *arbitrary* subset and $\Phi_j = (X_j, Y_j, Z_j)$ is a triple of rational functions of degree $\leq D$. It is *valid* if for every prime p, every j and every $v \in S_j$ with $Q(v) = \kappa p + \eta$, $\Phi_j(v) \in \mathbb{N}^3$ solves (1.1) for $n = p$. Put $R(p) = \bigcup_j \{v \in S_j : Q(v) = \kappa p + \eta\}$. A piece is *degenerate* if $X_j, Y_j, Z_j \in \mathbb{Q}(Q)$.

This covers every statement of the form "$\kappa p + \eta$ is represented by Q with side conditions, hence ES(p) by an explicit piecewise rational recipe" of fixed complexity. It does not cover recipes of unbounded algebraic complexity, or families with a parameter that grows with p.

Theorem 4.2 (PROVED, relative to Pila [Pi], Mordell–Schinzel [Mo, Sch] and [ET, Thm. 1.1]; [O124, Thm. 4.1, Cor. 4.2])**.** *Let Q be positive definite. There is a finite union U of primitive classes modulo some q^*, containing no square class, such that:*

(a) *for every large prime $p \notin U$, $\#R(p) \ll f(p)\, p^{(k-2)/2-1/4+\varepsilon}$, where $f(p)$ is the total number of solutions;*
(b) *for $0 < \varepsilon_0 < 1/4$ the primes $p \leq N$, $p \notin U$, with $\#R(p) \geq c p^{(k-2)/2-\varepsilon_0}$ number $\ll N^{3/4+\varepsilon_0+\varepsilon}$.*

Consequently, suppose a representation theorem (circle method for $k \geq 5$, Kloosterman/δ-method for $k = 4$, Duke [Du], Duke–Schulze-Pillot [DSP] and Blomer-type results for $k = 3$) gives $\#R(p) \geq c p^{(k-2)/2-\varepsilon_0}$ on a union $\mathcal{P}_R$ of primitive classes, outside $o(\pi(N))$ exceptions. Then $\mathcal{P}_R \subseteq U$: the representation input proves ES only for primes that a finite polynomial (Mordell-type) covering already handles.

Sketch. A degenerate piece produces an ES solution depending only on p. By a rationality and integrality argument it is a polynomial identity on finitely many classes, which by Mordell–Schinzel are not square classes; this defines U. A non-degenerate piece is non-constant on the quadric $Q = m$, $m = \kappa p + \eta$, outside finitely many p. Group the points of $R_j(p)$ by the value of a non-constant coordinate X_j; there are at most $f(p)$ values. Each group lies on a hypersurface section of the bounded quadric $Q = m$. For $k = 3$ such a section is a curve whose real components have degree ≥ 2 (no real line lies on a bounded quadric). By Pila's bound it has $\ll m^{1/4+\varepsilon}$ integer points in a box of side $\asymp m^{1/2}$. For $k \geq 4$ one slices by the last $k - 3$ coordinates. This gives (a). Then (b) follows from (a) and the first-moment bound $\sum_{p \leq N} f(p) \ll N \log^2 N \log\log N$ [ET, Thm. 1.1], and the corollary from Dirichlet's theorem. □

The same holds for line-free indefinite ternary forms [O124, §5]. The theorem does *not* say that an embedding proves ES only on U. A non-degenerate piece supported on a *thin* set can cover a positive proportion of primes. An example from [O124, §4]: $Q = \sum_{i \leq 5} v_i^2$, with $Q(v) = p + 1$, the piece $v_1 = 1$, $4 \mid Q(v)$, and the output $((p+1)/4 + v_1 - 1,\ p(p+1)/2,\ p(p+1)/2)$. What it excludes is that a representation *theorem*, which produces only sets of the generic size, does any work outside U. The only rank-3 quadratic piece of ES is Q_d (Proposition 2.10). It represents everything and is isotropic, and its side condition is not mild. EVIDENCE: every Mordell-hard prime $\leq 10^8$ is solved through Q_d with $d \leq 8$, $a \leq 22$ [O124, §8].

4.2. **The scale constraint.**

Definition 4.3 ([O127, Def. 2.2])**.** An *ES-detector* assigns to each large prime p a finite target $\mathcal{T}_p$ and a map $\Phi_p : \mathcal{T}_p \to \mathrm{Sol}(p)$ whose fibres have size $\le \mu(p)$. A hypothesis H *drives* it if H implies $\#\mathcal{T}_p \ge M(p) - R(p)$ with explicit $M(p) > R(p)$.

Proposition 4.4 (PROVED, relative to [ET, Thm. 1.1, Prop. 1.7]; [O127, Prop. 2.3], [RT1, Prop. 2.4, §5.5])**.** *If* H *drives a detector with* $R(p) \le M(p)/2$*, then* $M(p) \le 2\mu(p)f(p)$*. Here*

(a) $f(p) \le p^{3/5+O(1/\log\log p)}$ *for every* p [ET, Prop. 1.7]*;*
(b) *for each* $\delta > 0$*,* $f(p) \le p^{\delta}$ *outside a set of primes with counting function* $O_\delta(N^{1-\delta}\log^3 N)$*;*
(c) *for any* $\xi(p) \to \infty$*,* $f(p) \le \xi(p)\log^3 p \log\log p$ *outside a set of primes of relative density* $o(1)$*.*

In particular, if $\mu(p) \le p^{o(1)}$*, then* $M(p) \le p^{o(1)}$ *for almost all* p*, and no detector with* $M(p) \ge p^{\eta}$ *for all large* p *can be driven.*

Proof. $M/2 \le \#\mathcal{T}_p \le \mu(p)f(p)$. Part (a) is [ET, Prop. 1.7]. Parts (b) and (c) follow from Markov's inequality applied dyadically to $\sum_{p\le N} f(p) \ll N\log^2 N \log\log N$ [ET, Thm. 1.1]. □

Every natural parametrisation of solutions by lattice points, Heegner points or matrices is $O(1)$-to-one, or $p^{o(1)}$-to-one by the divisor bound. So any equidistribution statement whose main term for the target is a power of p is excluded as a driver, unless it detects only a coarse invariant of solutions (fibres $\mu \ge p^{\eta-o(1)}$). In that case the lifting step from the invariant to an integer solution must be supplied separately. This applies in particular to small-scale equidistribution of Heegner or CM points in families with $|D| \ge p^{\kappa}$ through a $p^{o(1)}$-to-one encoding [O127, Cor. 4.1]. With the Type I bound of [TH] in place of [ET, Thm. 1.1], Markov's inequality gives $f_I(p) \le \psi\mathcal{L}^3$ for all but a proportion $O(1/\psi)$ of primes [O123, Thm. 5.1]. A Planck-scale certificate must therefore have *polylogarithmic* accuracy.

Proposition 4.5 (local data of moderate modulus; PROVED; [O127, Prop. 7.3], [RT1, Lemma 5.1])**.** *Let* $S'(n)$ *count the solutions of the* $a = 1$ *Type I slice* ($t \ge 1$*,* $tn + 1 = mm''$*,* $m \equiv m'' \equiv -1 \pmod t$*,* $4 \mid m$*; then* $c = (m''+1)/t$*,* [O127, Lemma 1.1]*) with* $\gcd(n, c) = 1$*. Let* $\mathsf{H}(n)$ *be any predicate depending only on the dyadic range* $n \in (X, 2X]$ *and on* $n \bmod Q(X)$*, where* $Q(X) \le X^{1/2-\varepsilon}$*. If* H *holds at one large prime* $p \in (X, 2X]$ *that is a square modulo* $Q(X)$*, then it holds at an odd square in* $(X, 2X]$*, where* $S' = 0$*. Hence "*$\mathsf{H}(n) \Rightarrow S'(n) \ge 1$*" is false for any such* H*.*

Proof. The odd $m \equiv m_0 \pmod{2Q}$ with $m_0^2 \equiv p \pmod Q$ give $\asymp X^{1/2}/Q$ squares in the class of p. For odd $n = m^2$, a slice solution with $\gcd(n, c) = 1$ is a Type I solution, which cannot exist by [ET, Prop. 1.6]. □

This does not apply to "$\mathsf{H}(n) \Rightarrow \mathrm{ES}(n)$": odd squares satisfy ES, through solutions that are of neither type.

4.3. **Thickening a target is averaging over** n**.**

Proposition 4.6 (PROVED; [O123, Prop. 5.4])**.** *In the setting of Proposition 2.3, for* $z \in \Lambda_d$ *the integer* $k(z) = (2c\sqrt{d}\,x - \sqrt{d} - py)/y = 4acd - f - p$ *equals* $n - p$*, where*

$n = 4acd - f$ is the integer for which (a, c, d, f) satisfies Lemma 2.1(i) with p replaced by n. Hence the thickened target $R_c^{(K)} = \{|2c\sqrt{d}\,x - \sqrt{d} - py| \le Ky,\ y \ge \sqrt{d}/(2p)\}$ (the size window $f \le 2p$ of [O123, Prop. 3.1]*), of hyperbolic width $\asymp (2K+1)/p$ when $c\sqrt{d} \ll p$, counts exactly*

$$\sum_{|n-p|\le K} \#\{(a, f) :\ f \mid 4a^2d + 1,\ 4acd - f = n,\ f \le 2p\}.$$

Proof. The computation of Proposition 2.3, with $k(z) \in \mathbb{Z}$ allowed to range over $[-K, K]$. □

The counted points are Elsholtz–Tao points of the integers n near p. For composite n the coprimality $\gcd(c, n) = 1$ is not imposed, so the count is at least the Type I count of n with these (c, d), not equal to it. Assessment. A spectral estimate at Sobolev scale V (target width $\asymp 1/V$) therefore controls only averages of Type I counts over intervals $|n - p| \lesssim p/V$. With V a small power of the discriminant (§5) these are intervals of length $p^{1-\kappa}$. In that range unconditional exceptional-set bounds are already available, and nothing pointwise follows. The pointwise problem is exactly $K = 0$.

4.4. **Everything but one sign is soluble at odd squares.** In the lattice language of Proposition 2.5(c), Type II at (c, d) for an integer n asks for $Q = [A, B, C] \in L = \mathbb{Z}^3$ with $q(Q) = -4dn$, $b(Q, S_c) = 4$, $B \equiv 0 \pmod{4d}$, $d \mid C$ and $A > 0$.

Proposition 4.7 (proved; [O128, Prop. 8.3])**.** *For every $n \ge 1$, including odd squares, and every (c, d), the vector $Q_0 = [-1, 0, -dn]$ satisfies every condition above except $A > 0$: $q(Q_0) = -4dn$, $b(Q_0, S_c) = 4$, $B \equiv 0 \pmod{4d}$, $d \mid C$. Hence the Type II embedding problem is everywhere locally soluble, and even globally soluble without the sign condition, for every n. For odd squares $n = s^2$ it has no solution with $A > 0$ (Proposition 2.5(d)).*

Proof. $q(Q_0) = 0 - 4(-1)(-dn) = -4dn$ and, since $b(Q, S_c) = 4(cB - A)$, $b(Q_0, S_c) = 4$. Solubility at every prime ℓ is witnessed by Q_0. At ∞ take $B > 0$ large, $A = cB - 1 > 0$ and $C = (B^2 + 4dn)/(4A) > 0$. The last sentence is [ET, Prop. 1.6]. □

So no statement "everywhere locally soluble ⇒ soluble" holds for this integral-point problem. Q_0 is the trivial factorisation $N = (-1)(-N)$ of $N = 4c^2dn + 1$, i.e. $a = 0$. The content of Type II is that N has a positive divisor in the class $-1 \pmod{4cd}$, while it always has the positive divisor 1 in the class $+1$. The two differ only by the sign at ∞. The only restriction forced locally is the genus character $\chi_{-4d} \cdot (n/\cdot)$, a $\mathbb{Z}/2$ quotient that is non-trivial exactly at squares [O128, Prop. 9.2]. There is no natural spinor genus for this pair problem, and the phrase "spinor obstruction" does not apply. Once S_c is fixed, the stabiliser is a *split* torus, and the problem is representation by $x^2 - y^2$ in a coset, i.e. a divisor-in-class question [O128, §8]. This is the reason (an Assessment) why the Gauss/three-squares mechanism (class number one of a ternary genus, with an anisotropic stabiliser) does not transfer.

4.5. **Geometry: fibrations, strong approximation, Brauer–Manin.** Let U_n and α be as in Proposition 2.6, and let L_{ij} be the lines $u_i = u_j = 0$ and M_k the lines $u_k = n/4$, $u_i = -u_j$ (rational only if $4 \mid n$).

Theorem 4.8 (PROVED, relative to [BL, Lemma 3.10]; [O126, Thm. 3.1])**.** *Let $C \subset U_n$ be an irreducible curve over $\mathbb{Q}$ containing infinitely many integral points. Then C is one of the lines L_{ij}, or one of the M_k (only if $4 \mid n$; these carry no positive points). In particular, for every dominant map $U_n \to B$ to a curve, all fibres except finitely many lines have finitely many integral points, and no fibre has infinitely many positive integral points.*

Proof. Points with a zero coordinate lie on the L_{ij}. By [BL, Lemma 3.10], integral points with $u_1u_2u_3 \neq 0$ and $u_i \neq -u_j$ are finite in number. Points with $u_i = -u_j \neq 0$ satisfy $u_k = n/4$ and have a negative coordinate. □

The reason is the real place, not the splitness of the fibres: the open torus $V_n \subset \tilde{U}_n$ of [BL, Lemma 2.1] is split over $\mathbb{Q}$ [O126, Prop. 3.2]. No fibration argument can produce infinitely many positive points in a fibre.

Proposition 4.9 (PROVED; [O126, §1], [BL])**.** *For every $n \geq 2$, the statement "the Brauer–Manin obstruction is the only obstruction to positive integral points on U_n" is equivalent to* ES(n)*. For $n = 1$ the Brauer–Manin obstruction does obstruct.*

Sketch. For $n \geq 2$ the positive Brauer set is non-empty by [BL, Thm. 1.8], so "BM is the only obstruction" says exactly that a positive integral point exists. For $n = 1$ there is no positive solution, and every point of the positive adelic set has $\mathrm{inv}_\infty \alpha = -1$ and $\mathrm{inv}_\ell \alpha = +1$ for all finite ℓ [BL, Lemmas 3.1, 3.5, 3.8], so the Brauer set is empty [O126, §1]. (The proof of [BL, Thm. 1.8] needs a prime divisor of n; their Thm. 1.1 correctly assumes $n \geq 2$.) □

So a "BM is the only obstruction" conjecture for this family is ES restated.

On the homogeneous spaces $X_d : \ ef - 4da^2 = 1$ ($\cong \mathrm{Spin}(2,1)/T$, where strong approximation holds), the Type I locus is $\{ef - 4da^2 = 1,\ f + p = 4cda,\ c \neq 0\}$. The ES condition $4da \mid f + p$ is a congruence modulo a *coordinate of the point*, not modulo a fixed modulus. So it is not an adelic open condition away from ∞, and strong approximation does not reach it. For each (p, d) the locus has finitely many integral points, while $X_d(\mathbb{Z})$ is infinite [O126, Prop. 4.1] (PROVED). That descent and strong approximation are irrelevant beyond this is an ASSESSMENT [O126, Cor. 4.2]. The same shape holds for Type II on Q_d and on the level-dp shells.

4.6. **Single targets and fixed shells are empty infinitely often.**

Proposition 4.10 (CONDITIONAL on Hypothesis H; [O123, Prop. 5.3])**.** *There are infinitely many primes $p \equiv 5 \pmod{12}$ for which the genus-allowed Type I target $(c, d) = (1, 3)$ is empty, although its expected size is $\approx 0.17\,\mathcal{L}$.*

Proof. $x(x^2+12)$ on $x \equiv 5 \pmod{12}$ has no fixed prime divisor, so H gives infinitely many primes $p \equiv 5 \pmod{12}$ with $p^2 + 12$ prime. Here $\chi_{-12}(-p) = 1$. The target counts divisors f of $p^2 + 12$ with $f \equiv -p \equiv 7 \pmod{12}$ (Lemma 2.1(iii)). The only one is $f = 1 \not\equiv 7$. □

Proposition 4.11 (CONDITIONAL on Dickson's conjecture for two linear forms; [RT2, Lemma 1.2.1])**.** *For every fixed (c, d) there are infinitely many primes p for which the Type II shell (c, d) is empty, although its mean is $\asymp \mathcal{L}/\varphi(4cd)$.*

Proof. The pair $X,\ 4c^2dX + 1$ has no fixed prime divisor, so Dickson's conjecture gives infinitely many primes p with $4c^2dp + 1$ prime. Its divisors 1 and $4c^2dp + 1$ are $\equiv 1 \not\equiv -1 \pmod{4cd}$. □

Under H, applied to finitely many polynomials at once, any fixed finite union of such targets is empty infinitely often. This is Theorem 3.2(iii) seen from the fixed-p side. EVIDENCE: all 3630 primes $p \equiv 5 \pmod{12}$ in $[10^6, 3 \cdot 10^6)$ with $p^2 + 12$ prime have the $(1, 3)$ target empty, and among all genus-allowed primes near 10^6, 13–89% of the individual targets (c, d) are empty [O123, §3, §5].

5. Quantitative gaps to known equidistribution

Everything in this section is an ASSESSMENT: a comparison of exponents, not a theorem about what any method can do.

5.1. **What is known.** Young [Yo, Thm. 2.1] proves a hybrid subconvexity bound which gives, for Heegner points of discriminant $-D$, an error of the shape $\|\phi\|_2 D^{5/12+\varepsilon} V^{1+\varepsilon}$ for a test function at Sobolev scale V. Consequently every fixed ball of radius $D^{-\delta}$ with $\delta < 1/24$ receives its expected share of points, and $\delta < 1/8$ under the Lindelöf hypothesis; see also Humphries [Hu], who also gives the genus and closed-geodesic versions. Young's Prop. 1.2 is a *single-discriminant existence theorem* proved spectrally: $b^2 \equiv -D \pmod{4a}$ has solutions with $a \asymp \sqrt{D}$, $|b| \asymp \sqrt{D}/V$ for every large odd fundamental D, in the range $V \ll D^{1/18-\varepsilon}$. It is the model of what a spectral pointwise ES statement would have to look like [LA]. Humphries and Radziwiłł[HR] reach much smaller scales, but only for almost all centres. The ES targets are specific regions attached to cusps, not random ones.

5.2. **What ES needs.**

- *Type I.* In the T_p-translated picture the discriminant is $D' = 4dp^2$ and the target tubes have width $\asymp 1/p \asymp |D'|^{-1/2}$ (Proposition 2.3; [O123, Prop. 3.2]): exactly the Planck scale. In the level-d picture $D = 4d$ is tiny and the large parameter enters only through the shape of the target. Proposition 4.6 shows that thickening to a reachable width $|D'|^{-\delta}$ produces averages over n in intervals of length $p^{1-2\delta}$.
- *Type II.* The shells of Proposition 2.5(c) have width $\asymp D^{-1/2}/c$ with $D = 4dp$. Young's method reaches shells of width $D^{-1/18}$ unconditionally and $D^{-1/6}$ under the Lindelöf hypothesis [O128, §4]. The gap is a factor $D^{4/9}$, resp. $D^{1/3}$, so what is needed goes beyond GLH.
- *Accuracy.* By Proposition 4.4 and [O123, Thm. 5.1], the count to be certified is $\le \psi\mathcal{L}^3$ for all but a proportion $O(1/\psi)$ of primes. Termwise use of GLH or of the Linnik–Selberg conjecture loses p^{ε}, which exceeds every power of $\mathcal{L}$; individual central values are not polylogarithmically bounded. GRH can enter with only polylogarithmic losses in averaged form (moments, effective Siegel). But a Cauchy–Schwarz estimate over the spectrum at Planck scale still loses a factor $\approx p$ against the main term $\approx \mathcal{L}$ [O123, §5].
- *Termwise GRH on a character expansion.* Expanding the $a = 1$ slice count into Dirichlet characters and bounding each twisted divisor sum by a GRH-type estimate gives error $\ge p^{1-o(1)}$ against a main term $\asymp \log^2 p$ [O127, Prop. 9.1] (PROVED bookkeeping, mechanism scope).

Assessment 5.1. No standard hypothesis (GRH, GLH/Lindelöf in any aspect, Ramanujan–Petersson, Selberg's eigenvalue conjecture, twisted Linnik–Selberg) is known to suffice, by the routes of this note, to prove ES, or the Type I or Type II part of ES, for all large primes. A sufficient statement would have to be at the

Planck scale, of polylogarithmic accuracy, genus-aware (carrying $1+\chi_{-4d}(-p)$, without which it is false at $n=p^2$) and aggregated over a growing set of levels and cusps. Such a statement is ES restated, not a standard conjecture. Its level window must also be anchored at the least quadratic non-residue n_p. For $p\equiv 1 \pmod 4$, Type I at level d needs $(-d/p)=-1$, so a window $d\le D(p)$ forces $n_p\le D(p)$; GRH gives $n_p\ll\mathcal{L}^2$ (Ankeny, Bach) and Burgess gives $n_p\ll p^{1/(4\sqrt{e})+\varepsilon}$ unconditionally.

6. Two non-counting escapes that fail

Proposition 4.4 bounds every route that proves ES by counting a target against a main term. Two kinds of argument look different. A *covering* (pigeonhole) argument would show that a whole coset of residues is hit, hence also the target residue, without locating or counting solutions. A *fibred* encoding attaches to each solution a fibre that grows like a power of an auxiliary height H, so that the count is large although the number of solutions is small. This is the loophole "fibres $\mu\ge p^{\eta-o(1)}$" left open after Proposition 4.4 (the blind spot of the red team's test T8, [RT1, §5.5]). Both routes fail, each with an exact scope stated below. The sources are [LP] (covering) and [LF] (fibred), with their internal reviews.

6.1. **Covering in the e-slices.** Let p be an odd prime and $e\ge 1$ with $e\equiv -p \pmod 4$, $p\nmid e$. Put

$$N_e^{II}(p)=\frac{p+e}{4},\qquad N_e^{I}(p)=\frac{pe+1}{4},$$

and for $\gcd(N,e)=1$ define the *signed divisor-ratio set* $\Sigma_e(N)=\{ab^{-1} \bmod e : a,b\ge 1,\ ab\mid N\}\subset G_e:=(\mathbb{Z}/e)^*$.

Lemma 6.1 (e-slices; PROVED, elementary; [LP, Lemma 1.1])**.** *Let $T\in\{I,II\}$ and $N=N_e^T(p)$.* (i) $\gcd(N,e)=1$. (ii) *The elements of $V_T(p)$ with this value of e are in bijection with the pairs (a,b), $a,b\ge 1$, $ab\mid N$, $e\mid a+b$, via $d=N/(ab)$, $c=(a+b)/e$.* (iii) *So this fibre is non-empty iff $-1\in\Sigma_e(N)$.* (iv) $\Sigma_e(N)=\prod_{\ell^v\parallel N}\{\ell^t : |t|\le v\}$ *(a product set in G_e); in particular $|\Sigma_e(N)|\le\tau(N^2)$.*

Proof. (i) $4N^I\equiv 1$ and $4N^{II}\equiv p \pmod e$, and e is odd. (ii) These are the defining equations $4abd=4N$, $a+b=ce$. (iii) By (i), a and b are units mod e, and $e\mid a+b$ iff $ab^{-1}\equiv -1$. (iv) With $a=\prod\ell^{\alpha_\ell}$, $b=\prod\ell^{\beta_\ell}$ the condition is $\alpha_\ell+\beta_\ell\le v_\ell$, and every $t=\alpha-\beta\in[-v,v]$ occurs. □

Every element of $V_I(p)\cup V_{II}(p)$ has a value of e, so by (2.1), ES(p) holds iff $-1\in\Sigma_e(N_e^T(p))$ for some e and T: the e-slices parametrise all solutions by one coordinate. (The map from $V_T(p)$ to solutions is onto, and injective once $\gcd(a,b)=1$ is imposed; existence is unaffected.) For Type II, $N=N_e^{II}(p)$ runs over the consecutive integers above $p/4$.

From now on $p\equiv 1 \pmod 4$, so $e\equiv 3 \pmod 4$. Then the Kronecker symbol χ_{-e} coincides with the Jacobi symbol $(\cdot/e)$; it is a character of G_e with $\chi_{-e}(-1)=-1$. Let $K_e=\ker\chi_{-e}$, a subgroup of index 2 not containing -1.

Lemma 6.2 (genus inside a slice; PROVED; [LP, Lemma 1.2])**.** *Let $N=N_e^T(p)$.* (i) *Every prime $\ell\mid N$, including $\ell=2$, has $\chi_{-e}(\ell)=(\ell/p)$.* (ii) $\chi_{-e}(N^I)=1$ *and $\chi_{-e}(N^{II})=(e/p)$.* (iii) *Hence a point of the slice needs a prime $\ell\mid ab$ with $(\ell/p)=-1$. If p is replaced by an odd square m^2 with $\gcd(m,e)=1$, then $\Sigma_e(N)\subset K_e\not\ni -1$.*

Proof. (i) For odd ℓ, Type II gives $-e \equiv p$ and Type I gives $-e \equiv p^{-1} \pmod{\ell}$, so $(-e/\ell) = (p/\ell) = (\ell/p)$. If $2 \mid N$ then $8 \mid p+e$ (resp. $8 \mid pe+1$), so $-e \equiv p \pmod 8$, and $(-e/2) = (p/2) = (2/p)$. (ii) Multiplying (i), $\chi_{-e}(N) = (N/p)$, and $4N^{I} \equiv 1$, $4N^{II} \equiv e \pmod p$. (iii) $\chi_{-e}(ab^{-1}) = -1$ and (i); at m^2 all the symbols are $+1$. □

The target -1 lies in the non-trivial coset $-K_e$, and only prime factors with $(\ell/p) = -1$ leave K_e. This is the odd-square obstruction [ET, Prop. 1.6] inside one slice, and the reason why small levels fail when p is a residue of all small primes. The naive pigeonhole aims at the wrong coset:

Lemma 6.3 (collisions give signed solutions; PROVED; [LP, Lemma 1.3])**.** *If $2^{\omega(N)} > \varphi(e)$, there are coprime $a > b \geq 1$ with $ab \mid N$ and $a \equiv b \pmod e$. With $c = (a-b)/e$ and $d = N/(ab)$,*

$$\tfrac{4}{p} = \tfrac{1}{abd} - \tfrac{1}{acdp} + \tfrac{1}{bcdp} \quad (T = II), \qquad \tfrac{4}{p} = \tfrac{1}{abdp} - \tfrac{1}{acd} + \tfrac{1}{bcd} \quad (T = I).$$

Proof. Two of the $2^{\omega(N)}$ unitary divisors of N agree mod e; divide both by their gcd. The identities are $c(p+e)/(abcdp)$ and $c(1+pe)/(abcdp)$, as for (2.1). □

Collisions and zero-sum theory (Davenport, Olson, Erdős–Ginzburg–Ziv) hit the target $1 \in K_e$, i.e. signed solutions, which exist trivially. Positivity asks for a translate $-K_e$ to be hit: a covering statement.

Proposition 6.4 (covering criterion; PROVED from Kneser's theorem [Kn], [TV, Thm. 5.5]; [LP, Prop. 1.4])**.** *Let G be a finite abelian group, $g_1, \dots, g_k \in G$, $v_i \geq 1$, $A_i = \{g_i^t : |t| \leq v_i\}$, $\Sigma = A_1 \cdots A_k$, $t_0 \in G$, and let $o_H(g)$ be the order of gH in G/H. If*

$$\textstyle\sum_i \big(\min(2v_i+1, o_H(g_i)) - 1\big) \;\geq\; |G:H| - 1$$

for every subgroup H with $t_0 \notin H$, then $t_0 \in \Sigma$. Among criteria that depend only on the orders $o_H(g_i)$ (for all H) and on the v_i, this one is sharp: in $\mathbb{Z}/2m$ with all $g_i = 1$, $v_i = 1$, $t_0 = m$ it reads $2k \geq 2m-1$ at $H = 0$ (and is implied by it at the other H), and $\Sigma = [-k, k]$ contains m iff $k \geq m$.

Proof. Let $H = \mathrm{Stab}(\Sigma)$. As $1 \in \Sigma$, $H \subset \Sigma$; if $t_0 \notin \Sigma$ then $t_0 \notin H$ and $|\Sigma/H| \leq |G:H| - 1$. In G/H every partial product of the images X_i of the A_i has trivial stabiliser, so Kneser gives $|\Sigma/H| \geq 1 + \sum_i (|X_i| - 1)$ with $|X_i| = \min(2v_i+1, o_H(g_i))$. □

In the slice ($G = G_e$, g_i the primes $\ell \mid N$, $v_i = v_\ell(N)$, $t_0 = -1$), $H = K_e$ recovers Lemma 6.2, and $H = \{1\}$ requires $\Omega(N) \geq (\varphi(e) - 1)/2$. So there are three regimes. Counting: a certificate that $-K_e \subset \Sigma_e(N)$ needs $\tau(N^2) \geq \varphi(e)/2$, i.e. $\Omega(N) \geq \log_3(\varphi(e)/2)$. Structure-only criteria as above need $\Omega(N)$ linear in $\varphi(e)$. Random residues would need only $3^{\omega(N)} \gtrsim \varphi(e)\log\varphi(e)$ (a heuristic). To go from linear to logarithmic one must know the actual residues $\ell \bmod e$ of the prime factors. Forcing small primes into N by congruences costs the modulus: $\ell \mid N_e^{II}(p)$ iff $e \equiv -p \pmod{4\ell}$, so forcing k odd primes puts e in one class modulo $4L \geq \exp((1+o(1))k\log k)$, against the 3^{k+1} residues available [LP, Lemma 2.1]. And by the Nicolas–Robin bound [NR], any coset of index $\leq e^{\theta}$ ($\theta < 1$) contained in $\Sigma_e(N_e^T(p))$ forces $e \leq \exp(O_\theta(\log p/\log\log p))$ [LP, Lemma 2.3]: covering certificates live in a window of length $p^{o(1)}$.

The primes of N below the modulus have known residues (themselves). The next theorem shows that they never suffice on a positive proportion of primes.

Definition 6.5 ([LP, §2.2])**.** Let $N = N_e^T(p)$, $L = \prod_{\ell^v \| N,\ \ell < e} \ell^v$ (the *transparent part*), $M = N/L$, and $R = \Sigma_e(L) \cdot \{M^{-1}, 1, M\}$. For $0 \le \theta < 1$, a *transparent covering certificate of strength* θ at (p, e, T) is a subgroup $H \le G_e$ with $|G_e : H| \le 2e^\theta$ and $-H \subset R$.

Since $R \subset \Sigma_e(N)$ and $-1 \in -H$, a certificate yields a solution. With $H = K_e$ it says that R covers the whole target coset $-K_e$.

Theorem 6.6 (PROVED, relative to PNT in progressions to a fixed modulus, Brun–Titchmarsh [MV], Shiu [Sh, Thm. 1] and Nicolas–Robin [NR]; [LP, Thm. 2.2])**.** *Fix $0 \le \theta < 1/2$ and $\varepsilon > 0$. There is $C = C(\theta, \varepsilon)$ such that for every $Y \ge 3$, with $q_0 = 8 \prod_{3 \le \ell \le Y} \ell$,*

$$\#\{p \le x :\ p \equiv 1\ (q_0),\ \text{a certificate of strength } \theta \text{ exists at some } (e, T)\} \le \big(CY^{-1+2\theta+\varepsilon} + o_{x\to\infty}(1)\big)\, \pi(x; q_0, 1),$$

where e ranges over all $e \equiv 3 \pmod 4$ and T over $\{I, II\}$. Hence, for Y large, a set of primes of positive lower density has no transparent covering certificate in any e-slice of either type.

Sketch. A certificate gives $|R| \ge \varphi(e)e^{-\theta}/2$, hence $\tau(L^2) \ge B_e := \varphi(e)e^{-\theta}/6$. Fix $p \equiv 1 \pmod{q_0}$, $p \le x$.

(a) $e \le Y$. Every prime $\ell < e$ has $(\ell/p) = 1$, so $L \in K_e$ (Lemma 6.2(i)); $(e/p) = 1$ as all prime factors of e are $\le Y$, so $\chi_{-e}(M) = 1$ by Lemma 6.2(ii). Thus $R \subset K_e \not\ni -1$: not even a transparent witness exists.

(b) $Y < e \le E_1 := \exp((\log x)^{1/3})$. If $L > x^{1/2}$, multiplying the prime factors of L (all $< x^{1/4}$) one at a time gives $u \mid N$ with $x^{1/4} < u \le x^{1/2}$ and $P^+(u) < E_1$; then p lies in one class mod $4u$, and Brun–Titchmarsh with Rankin's bound for $\sum_{u > x^{1/4},\ P^+(u) < E_1} 1/\varphi(u)$, summed over the $\le 2E_1$ pairs (e, T), gives $o(x/(\varphi(q_0)\log x))$ primes. If $L \le x^{1/2}$, use Markov's inequality with the second moment: $1 \le B_e^{-2}\tau(L^2)^2 = B_e^{-2}\sum_{m \mid L} h(m)$, where $h = \tau(\cdot^2)^2 * \mu \ge 0$, $h(\ell^v) = 8v$. Each $m \le x^{1/2}$ with $m \mid N$ puts p in one class mod $\operatorname{lcm}(4m, q_0) \le x^{2/3}$ (compatible with $p \equiv 1\ (q_0)$ only if every odd $\ell \le Y$ dividing m divides $e + 1$). Brun–Titchmarsh and an Euler product bound the number of such p by $\ll \frac{x}{\varphi(q_0)\log x}\, 19^{\omega(e+1)}(\log e)^8 B_e^{-2}$, and $\sum_{e > Y} 19^{\omega(e+1)}(\log e)^8 B_e^{-2} \ll Y^{-1+2\theta+\varepsilon}$ with a constant independent of Y.

(c) $E_1 < e \le x$. Markov over all $p \le x$ (the map $p \mapsto N$ is injective): Type II has $N \le x/2$ and $\sum_{n \le x/2} \tau(n^2)^2 \ll x(\log x)^8$; Type I has $N \equiv 4^{-1} \pmod e$, $N \le X = (xe + 1)/4$, and Shiu's theorem (modulus $e < X^{3/4}$) gives $\ll (xe/\varphi(e))(\log x)^8$. Since $\sum_{e > E_1} B_e^{-2} e/\varphi(e) \ll E_1^{-1+2\theta+\varepsilon}$, which beats every power of $\log x$, the count is $o(x/(\varphi(q_0)\log x))$.

(d) $e > x$. Then $N \le e^2/4$, and Nicolas–Robin gives $\tau(N^2) \le \tau(N)^2 \le \exp(4.3 \log e/\log\log e) < B_e$ for $e \ge e_0(\theta)$.

Finally, PNT in progressions gives $\pi(x; q_0, 1) \ge x/(2\varphi(q_0)\log x)$ for $x \ge x_0(Y)$. □

Remark 6.7 (all $\theta < 1$)**.** The restriction $\theta < 1/2$ comes only from the second moment. With $\tau(\cdot^2)^r$, $r > 1/(1-\theta)$, the same four steps go through (local factors $\ll_r v^{r-1}$, Shiu with $(3^r)^v$, Markov with B_e^{-r}), so the theorem holds for every fixed $\theta < 1$: transparent certificates covering a coset of size $\ge e^\delta$ fail on a positive-density

set, for every $\delta > 0$. This extension was derived by the internal reviewer and checked step by step against the proof above (`reviews/longshot-pigeonhole-review.md`, D6); it is recorded in [LP, §2.2] as a remark.

Scope. Theorem 6.6 concerns certificates that use only the prime factors of $N_e^T(p)$ below the modulus, plus the rest of N as one opaque block, and that cover a coset of size $\geq e^\delta$. It says nothing about arguments that merely hit -1, or that use the residues mod e of specific prime factors above the modulus. Those are the per-slice divisor events of §3, i.e. hatches H-A and H-B below. The proof shows more than failure: the expected number of transparent certificates over all e is bounded by a convergent series $\sum_e (\log e)^8 e^{2\theta-2}$ independent of p, so this is a bounded-mass mechanism, not a near miss. The analogous statement for the two-coordinate rows of Lemma 3.1 should follow by the same proof (ASSESSMENT; not written out).

Conjecture 6.8 (PH, CONJECTURE; [LP, Conj. 5.1])**.** For every sufficiently large prime $p \equiv 1 \pmod 4$ there are $e \equiv 3 \pmod 4$ and T with $\varphi(e) \geq 40$ and $\Sigma_e(N_e^T(p)) \supseteq -K_e$.

PH implies ES(p). EVIDENCE [LP, §3] (independent engine for six primes in the review): every Mordell-hard prime below 10^6 except 3529 and 110881 has such a cover with $e \leq 2000$; so do 300 consecutive Mordell-hard primes after each of $10^6, 10^7, 10^8$ (with $e \leq 5000$), and 15 selected hard primes (with $e \leq 20000$). The largest covered cosets have $|-K_e|$ up to 2339, and the deep covers are almost always transparent. For $e \geq 500$, covers are about 5% of the slices containing a solution at 10^8: most solutions at moderate e hit -1 without covering $-K_e$. The transparent counts still grow up to 10^8, so the failure set of Theorem 6.6 is not visible numerically; the theorem is qualitative. By Theorem 6.6, any proof of PH must use prime factors above the modulus on a set of positive density, so PH is not easier than H-A/H-B (ASSESSMENT).

6.2. **Fibred encodings.** Write Sol(n) for the (finite) set of solutions of (1.1) and $f(n)$ for the number of Type I and Type II solutions. A *fibred encoding* at n is a countable set $E = E(n)$ with a height $\mathrm{ht} : E \to [0, \infty)$ ($E(\leq H)$ finite) and a map $\varphi : E \to \mathrm{Sol}(n)$. It is *power-fibred* if $\#E(\leq H) = c(n)H^\delta(1 + o(1))$ as $H \to \infty$, for some $\delta > 0$. A *counting route* proves this asymptotic with an independent formula for $c(n)$ and reads off $c(n) > 0 \Rightarrow \mathrm{Sol}(n) \neq \emptyset$. If each fibre $E_s = \varphi^{-1}(s)$ has a density c_s, then $c(n) = \sum_{s \in \mathrm{Sol}(n)} c_s$, a finite sum [LF, Lemma 1.2]. The H-asymptotic is free; all the content sits in the constant. The question is whether $c(n)$ can be evaluated without knowing Sol(n), and that depends on how φ is computed.

Lemma 6.9 (rigidity; PROVED, elementary; [LF, Lemma 2.1])**.** *Let $V/\mathbb{Q}$ be irreducible, $E \subset V(\mathbb{Q})$ Zariski-dense, and $\varphi : V \dashrightarrow \mathbb{A}^3$ a rational map over $\mathbb{Q}$, regular on E, with $\varphi(E)$ finite. Then φ is constant.*

Proof. On $U = \mathrm{dom}\,\varphi$ (irreducible), E is dense and covered by the finitely many closed sets $\varphi^{-1}(s_i)$; so one of them is U. □

By Borel's density theorem, an orbit Γx of an arithmetic subgroup Γ of a connected semisimple $\mathbb{Q}$-group without compact factors is Zariski-dense in $\overline{Gx}$ [LF, Lemma 2.2]. Hence every orbit of the kind counted by Duke–Rudnick–Sarnak [DRS] or Eskin–McMullen [EM] is covered by Lemma 6.9. Consequences (PROVED, [LF, Thm. 2.3]): if E is a union of pieces E_i, each Zariski-dense in an irreducible V_i

with $\varphi|_{E_i}$ rational, then $\varphi \equiv s_i$ on E_i, and s_i is the value of φ_i at any point of its domain, so it can be read off the equations. If moreover φ_i maps V_i into the ES variety $4xyz = n(xy + yz + zx)$ and has $\mathbb{Z}_\ell$-coefficients, then one $\mathbb{Z}_\ell$-point for each ℓ and one real point in the counting region already force $s_i \in \mathrm{Sol}(n)$. A Siegel or Hardy–Littlewood main term carries no information beyond the explicit point s_i: for algebraic recipes counting is idle. (This holds for pieces as given. A candidate set $V_i(\mathbb{Z}) \cap \Omega$ that is not Zariski-dense must first be decomposed along the components of its closure.)

Corollary 6.10 (PROVED, relative to [ET, Prop. 1.6] and Dirichlet; [LF, Cor. 2.4])**.** *Let $\mathcal{V} \to \mathbb{A}^1_n$ be an irreducible $\mathbb{Q}$-family with geometrically irreducible generic fibre, and $\varphi : \mathcal{V} \dashrightarrow \mathbb{A}^3$ rational. Let P_{cov} be the set of primes p for which a Zariski-dense $E(p) \subset \mathcal{V}_p(\mathbb{Z})$ has $\varphi(E(p)) \subset \mathrm{Sol}(p)$. There is $D' \geq 1$ such that all but finitely many $p \in P_{\mathrm{cov}}$ are quadratic non-residues mod D'. In particular all large $p \equiv 1 \pmod{D'}$ are missed.*

Sketch. By Lemma 6.9 and generic flatness, φ is constant on the fibres over a dense open set, given by a point $s_0 \in \mathbb{Q}(n)^3$ on the ES variety; integrality at infinitely many primes forces $s_0 \in \mathbb{Q}[n]^3$. Fix the Type I or Type II pattern occurring infinitely often; e.g. for Type I, $x = nx_1$ and $y(0)z(0) \neq 0$. If p is a square mod D' (built from the denominators and $y(0)z(0)$), choose by Dirichlet a large prime $m \equiv m_0$ with $m_0^2 \equiv p \pmod{D'}$; then $s_0(m^2)$ is a Type I solution for the odd square m^2, contradicting [ET, Prop. 1.6]. □

This is the classical observation that polynomial identities miss the square classes ([ET, Props. 1.8–1.9]; Mordell–Schinzel), now for algebraic fibred encodings of bounded complexity, however large the fibres. With a parameter set growing with p the fibring changes only the normalisation of the sum over the parameter [LF, Cor. 2.5].

The remaining power-fibred encodings use *orbit recipes*: $E(n)$ is a finite union of Γ'-orbits in $V(\mathbb{Z}) \cap \Omega$, where $V = \overline{Gx_0}$ is a closed orbit of a connected semisimple $\mathbb{Q}$-group G with $G(\mathbb{R})$ without compact factors, $\Gamma' \subset G(\mathbb{Z})$ is arithmetic, and φ is Γ'-invariant. Let $\mu(n)$ be the largest number of orbits over one solution. The *counting input* (C) is the asymptotic $\#(\Gamma' y \cap B_H) \sim w_O \operatorname{vol}(B_H)$ for well-rounded B_H, with $w_O = \operatorname{vol}(\Gamma'_y \backslash H_y(\mathbb{R}))/\operatorname{vol}(\Gamma' \backslash G(\mathbb{R}))$, H_y the stabiliser [DRS, EM, EMS]. The statements of these counting theorems (and of [BR, CTX] below) were checked only second-hand, by a research subagent of the source; the internal reviewer had no access to them.

Theorem 6.11 (orbit weights; PROVED relative to (C); [LF, Thm. 3.1])**.** (a) $c(n) = \sum_O w_O = \sum_{s \in \mathrm{Sol}(n)} \sum_{O \mapsto s} w_O \leq \mu(n) f(n) \max_O w_O$. (b) *If all stabilisers $H_y(\mathbb{R})$ are compact, then $w_O = \kappa/|\Gamma'_y|$ with κ depending only on the real orbit of y, and $w_{\max}/w_{\min} \leq C(\Gamma')$. So the number of orbits is at most $C(\Gamma')\mu(n)f(n)$, and Proposition 4.4 holds with multiplicity $C(\Gamma')\mu(n)$.* (c) *Only non-compact stabilisers (tori or semisimple) can give weights that vary by a power of n.*

Proof. (a) is $c(n) = \sum_s c_s$ with (C) for each orbit. (b) Compact stabilisers in one real orbit are conjugate, so they have equal volume; Γ'_y is finite, and $G(\mathbb{Z})$ has finitely many conjugacy classes of finite subgroups. □

Examples [LF, §4]. The Type II pair embeddings (Q, S_c) of Proposition 2.5(c), moved by $\mathrm{SL}_2(\mathbb{Z})$, have stabiliser $\{\pm I\}$; each orbit contributes $\asymp H$ points of height

$\le H$, and $c(n) = \kappa_{c,d} \cdot \#\{f \mid 4c^2dn+1 :\ f \equiv -1\ (4cd)\}$. Numerically every orbit, ES or not, has the same count to within 1.5% at $H = 3000$ (EVIDENCE). The fibre is a copy of $\mathrm{PSL}_2(\mathbb{Z})$ and carries no information. The Type II Pell orbits $\{\varepsilon_s^k\}$ grow only logarithmically, contain no ES point after $k = 1$, and have regulators $O(\log p)$, because the solution is the unit; the Type I hyperboloids X_d give weights bounded above by a constant. The quaternion encodings have finite unit groups.

The odd-square test. Suppose a main term has the Siegel or Borovoi–Rudnick shape [BR] $M(n) = m_\infty(n)\prod_\ell m_\ell(n) \cdot |B|^{-1}\sum_{\alpha\in B}\prod_\ell E_\ell(\alpha, n)$ (local densities, and local Brauer averages E_ℓ for a finite $B \subset \mathrm{Br}$), subject to (U): there are a finite set S_0 of primes and $k \ge 1$ such that for $\ell \nmid n$, $m_\ell(n) > 0$ and $E_\ell = 1$ if $\ell \notin S_0$, while for $\ell \in S_0$ these factors depend only on n mod ℓ^k; and $m_\infty(n) > 0$ for large n.

Proposition 6.12 (PROVED under (U); [LF, §5.1])**.** *Let $Q_1 = \prod_{\ell\in S_0}\ell^k$, $p \nmid Q_1$ prime with $M(p) > 0$, and m odd, $\gcd(m, Q_1) = 1$, $m \equiv m_0 \pmod{Q_1}$ with $m_0^2 \equiv p$. If $M(m^2) = 0$, then the two values are separated at a place $\ell \mid m$ (some $m_\ell(m^2) = 0$, or some $E_\ell(\alpha, m^2) \ne 1$), or at the place p alone.*

Proof. By (U), every other factor takes the same value at p and at m^2. □

So a main term can pass the odd-square test only through data at the primes dividing m, which for $n = p$ means the place p. Such factors do occur: the spinor exceptions of $3x^2 + 4y^2 + 9z^2$ are exactly the squares w^2 with all prime factors of $w \equiv 1 \pmod 3$ (Earnest [Ea, Prop. 5.1]; reproduced numerically to $5 \cdot 10^4$). The ES-natural candidate is the genus set $\mathfrak{G}_d^-(n)$ of primitive positive forms of discriminant $-4dn$ with $\chi_{-4d}(Q) = -1$ (evaluated at any represented integer prime to $4dn$). The Type II form $[f, 4ad, de]$ of Proposition 2.5(c) lies in it (part (d)).

Proposition 6.13 (PROVED; [LF, §5.3])**.** (a) *For odd $n = m^2$ with $\gcd(m, 2d) = 1$, $\mathfrak{G}_d^-(m^2) = \emptyset$.* (b) *For a prime $p \equiv 1 \pmod 4$, $p \nmid 2d$, $\#\mathfrak{G}_d^-(p) = h(-4dp)/2$.*

Proof. (a) A represented k prime to $4dm^2$ has $1 = (-4dm^2/k) = \chi_{-4d}(k)$. (b) On represented values $\chi_{-4d} = (\cdot/p)$, a genus character which is non-trivial on the class group (there is a second assigned character, at 2 or at a prime $q \equiv 3 \pmod 4$ dividing d), so it takes the value -1 on half of the classes. □

The mass of $\mathfrak{G}_d^-$ therefore passes the odd-square test, through the place p, as Proposition 6.12 requires. But it is not a valid encoding: it has $h/2 \asymp p^{1/2\pm\varepsilon}$ classes against $F_d(p) = \#\{\text{Type II points at level } d\} = O(\text{polylog})$ on average. EVIDENCE: for 5 primes $p \equiv 1 \pmod 4$ in $[10^3, 10^4]$, chosen in the square classes mod 120 where Type II is sparsest, and $d \le 8$, $\#\mathfrak{G}_d^-(p) \in [10, 144]$ while $F_d(p) = 0$ in 19 of the 40 pairs (reproduced exactly by the reviewer; not a typical rate: $F_1(p) > 0$ for 55 of 60 consecutive primes $p \equiv 1 \pmod 4$ from 10^3). So the per-level criterion "$\mathfrak{G}_d^-(p) \ne \emptyset \Rightarrow F_d(p) > 0$" is false. The aggregate criterion "$\exists d$: $\mathfrak{G}_d^-(p) \ne \emptyset$" is not false but vacuous: by (b) it holds at every $p \equiv 1 \pmod 4$, so as a criterion for ES it is ES itself. Likewise the genus-full closure of the pair embeddings contains Q_0 for every n (Proposition 4.7), so its mass is positive at odd squares.

Theorem 6.14 (fibred encodings, exact scope; [LF, Thm. 5.4])**.** *Let $E(n) = \bigsqcup_{\theta\in\Theta(n)} E_\theta$ be power-fibred, with $\Theta(n)$ arbitrary (it may grow with n), each piece either* (X1) *Zariski-dense in an irreducible V_θ with $\varphi|_{E_\theta}$ rational, or* (X2) *a finite union of orbits as above, with orbit weights in $[w_\theta, Rw_\theta]$. Let $\mu(n)$ be the largest number of pieces (X1 pieces, X2 orbits) over one solution.*

(1) (PROVED) *X1 pieces are constant; X2 pieces satisfy Theorem 6.11, with* $R = C(\Gamma_\theta)$ *for compact or finite stabilisers and* $R = O(\log n)$ *for the Pell orbits.*
(2) (PROVED RELATIVE TO (C) AND [ET, Thm. 1.1]) *A counting route proving* $c(n) \geq M(n)/2$ *needs* $M(n) \leq 2R\mu(n)f(n)\max_\theta w_\theta$. *After normalising by* $\max_\theta w_\theta$, *Proposition 4.4 holds with multiplicity* $R\mu(n)$. *The factor* H^δ *cancels.*
(3) (CONDITIONAL *on the hypotheses of Colliot-Thélène–Xu* [CTX, Thm. 3.7]*:* G *simply connected, stabiliser connected*) *A main term from a mass formula is the mass of a genus-full set; if that set is the encoding, its non-emptiness is decided by local and Brauer–Manin data without counting, and by Proposition 6.12 the vanishing at odd squares must come from the places dividing* m. *These hypotheses fail for the pair space of Proposition 2.5(c) (*$G = \mathrm{SL}_2$, *stabiliser* μ_2*). The refutations of the two ES-natural genus-full criteria above do not use* [CTX].
(4) (PROVED) *If* Θ *is finite and the X1 data are uniform in* n *as in Corollary 6.10, all large* $p \equiv 1 \pmod{D'}$ *are missed.*

Exact scope. Theorem 6.14 covers algebraic recipes and orbit recipes with bounded covolume ratio R. It shows that the *continuous* fibre factor H^δ buys nothing. It is not a theorem about all large-fibre encodings. The red team's T8 blind spot is closed only partially: it moves from the size of the fibres to (E-d) the number $\mu(n) \geq n^\eta$ of pieces over one solution (discrete multiplicity; Theorem 6.14 is then vacuous) and to (E-a) weight spread $R \geq n^\eta$ not capped by the solution (none found: the only split torus among ES encodings is the Pell unit, which is the solution itself). Also outside are (E-b) a *valid* genus-full criterion with a Brauer or spinor factor at p, which would be a structural theorem of the strength of ES, and (E-c) recipes that are neither algebraic nor lattice-invariant (floors, gcds, factorisations), which fall under [PS, Thm. M] instead. Bilinear sieves fibred over free parameters remain single-target counts with polylogarithmic mass (ASSESSMENT).

7. OPEN PROBLEMS: THE ESCAPE HATCHES

By Theorem 3.2(iii), a route that survives must use a parameter set growing with p, integers that are not formal polynomials in p, non-abelian data, or a signed counting certificate. We list the concrete forms [RT2, §3]. The verdicts are ASSESSMENTS. Each hatch is equally a hatch for *disproof*: a sterile prime, if one exists, would be a failure not explained by Brauer–Manin (Proposition 4.9), and none of the encodings above could detect it either.

Problem 1 (H-A: joint factorisation of a growing family)**.** Prove, at a single large prime p, a lower bound for $\sum_{\theta\in\Theta(p)} \#\{f \mid N_\theta(p) :\ f \bmod q_\theta \in R_\theta(p)\}$ for a family $\Theta(p)$ growing with p; for example $\sum_{a\leq A,\, d\leq D} \#\{f \mid p + 4a^2d :\ f \equiv -1 \pmod{4ad}\}$ (row II$\{a, d\}$), or the Type I family $p^2 + 4c^2d$.

Divisor sums over the values of a *fixed* quadratic polynomial are classical [Ho], and roots of quadratic congruences are equidistributed on average over the modulus [DFI]. We know no result uniform in the constant term p in the needed range. More seriously, ES needs a *moving* modulus $q = 4ad$ tied to the variable. Once $q > N^{1/3}$, at most 11 divisors of N lie in one class [Le], so each θ contributes $O(1)$. The family is a sum of $\asymp \mathcal{L}^3$ events of size $O(1)$, with no large sub-family (§8). *Verdict:*

open; no known technology. This is the main hatch, and any automorphic route (Kuznetsov over the level) lives here, with the polylogarithmic accuracy demanded by Proposition 4.4.

Problem 2 (H-B: parity-breaking for a divisor-in-moving-class event)**.** A slice is empty when $N_\theta(p)$ is prime (Propositions 4.10, 4.11). More generally it is empty when all prime factors of $N_\theta(p)$ lie in classes generating a subgroup that misses $R_\theta(p)$. Exclude, at a fixed p, the conspiracy in which every $N_\theta(p)$, $\theta \in \Theta(p)$, is bad in this sense.

"Bad" is a multiplicative-class condition of parity type, which lower-bound sieves cannot detect without bilinear input [FI, ch. 16]. Parity-breaking through bilinear sums exists for thin polynomial sequences averaged over their variables [FI98, HB, Ma]. At fixed p the variables θ are free to average over. The differences are the target (a divisor in a moving class, not primality) and the size of the family ($\asymp \mathcal{L}^3$ effective members). *Verdict:* open; the closest existing technology in spirit. The best-defined non-automorphic research target in this list is a "Friedlander–Iwaniec theorem for divisors in moving classes over the Elsholtz–Tao family at fixed p". The $O(1)$-per-block mass makes bilinear methods, which need long sums, hard to apply.

Problem 3 (H-D: non-abelian or higher-rank data)**.** Find an object carrying information not determined by congruences (Frobenius in non-abelian extensions, GL_3 or higher automorphic forms) with non-zero coefficient in an ES count.

By Proposition 2.9(iii), in the Type I CM torus the dihedral data have coefficient 0, so such data could only enter through the choice of θ, not by deciding a fibre. The varieties are toric or quadric (Lemma 3.1), so no natural higher-rank object is attached. *Verdict:* open, with no candidate object known.

Problem 4 (H-E: an exact identity with positive main term)**.** Find one expansion of an aggregated ES count, with a positive main term and cancellation between different θ that is not visible piecewise.

Termwise use of all standard hypotheses is excluded at the level of mechanism (§5); even the Linnik–Selberg conjecture for sums of Kloosterman sums (partial results: [ST]) loses p^{ε}. The search of [O131] found no structurally positive identity. Its exact content: the Type II count $T_2^+(n)$ of [O131] (positive a, b, no coprimality conditions) vanishes at every square $n = m^2$, odd or even; for $cd \mid 6$ the Type II (c, d)-slice is an exact combination of binary theta coefficients at $4c^2dn + 1$; the Type II count is "Eisenstein even/odd parts minus a boundary -2 per level" [O131, Prop. 4.1]; and there is no linear relation between 56 ES count variants and 54 classical arithmetic functions on $n \le 2 \cdot 10^4$ (CERTIFIED, exact $\mathbb{Q}$-rank; the scope is that finite dictionary). In every case positivity is again "a specific integer has a divisor in a residue class". *Verdict:* open.

Remark 7.1 (H-F: integral Hasse principles, a cautionary example)**.** ES(p) asks for an integral point on an affine variety whose expected number of points is polylogarithmic. In the closest family where the integral Hasse principle has been settled, the Markoff-type surfaces, it holds for almost all parameters but fails for infinitely many [GS, Thm. 1.2]. In that setting "almost all" is what geometry delivered, not "all".

Remark 7.2 (non-escapes)**.** Anchoring the levels at the least non-residue n_p (H-C) fixes the genus condition but leaves H-A at the anchored levels. Taking Type I and Type II together (H-G) has the same $\mathcal{L}^3$ mass structure, and Theorem 3.2(iii) applies to unions of rows of both types [RT2, §3].

8. Numerical evidence

All statements here are EVIDENCE. The counts are exact enumerations, validated against brute force for $p \le 400$ and reproduced independently by internal reviewers, as indicated.

- $f_I(p) \ge 1$ and $f_{II}(p) \ge 1$ for every prime $5 \le p \le 10^8$, and the $a = 1$ Type I slice is non-empty for each of them [O123, §1], [O128, §10]. This was reproduced to 10^7 (Type I) and to 10^8 (Type II) by independent code.
- Dyadic minima of f_I grow with p and stay at 0.12–0.20 times the dyadic mean. The mean is $\approx 0.1\mathcal{L}^3$. For example $\min f_I = 105$ on $[2^{25}, 2^{26})$, at $p = 37117609$. All window minimisers from 2^9 on are Mordell-hard (in the six square classes modulo 840). The $a = 1$ slice alone has minimum 1–5 throughout.
- $\min f_{II}/\log^3 p \approx 0.012$ over $p \ge 10^5$, attained at $p = 2031121$; dyadic minima of f_{II} are $62, 74, 88, 94$ on the blocks $2^{23}, \dots, 2^{26}$ (the last one partial). The hardest primes lie in the square classes modulo 840; the 20 smallest normalised values are all $\equiv 1 \pmod{24}$ except one.
- Single fixed targets are often empty: the $d = 1$ Type II family (the single discriminant $-4p$) is empty for 148–198 primes per block at 2^{24}–2^{26}. Among primes $p \equiv 1 \pmod{24}$, $p < 10^5$, with $(-d/p) = -1$, Type I at the single level $d = 7$, 11, 13 fails for 42/612, 23/611, 88/608 primes, while $d = 5$ never fails. It is an "identity level": a classical polynomial covering [O126, §5.3]. Type II at level $d = 1$ fails for 73 of the 14183 primes $p \equiv 1 \pmod{24}$ below $1.5 \cdot 10^6$.
- The slice of [O127, Lemma 1.1] solves every Mordell-hard prime below 10^7, but its count drops to 1 at $p = 5101441$, where $f_I = 188$ and $f_{II} = 78$.
- Type II solutions near 10^6 sit at about one per dyadic (a, c, d) block [RT2, §3.1]: the mass is spread over $\asymp \mathcal{L}^3$ small events, with no large sub-family.

9. Summary

The automorphic and geometric objects naturally attached to (1.1) at a fixed prime (SL_2 matrices and Heegner points, T_p translates, shells, quaternion orders, CM tori, an isotropic ternary quadric, the Brauer class) are exact reformulations. For fixed p each of them reduces to genus or congruence data plus divisor events of specific integers at the Planck scale, with polylogarithmic expected counts (Theorem 3.2). Assessment: by these routes and with known technology, no standard hypothesis is known to suffice. What remains open is the joint control of a growing family of divisor events at one prime (H-A, H-B), or a genuinely new object (H-D, H-E). The Erdős–Straus conjecture remains open.

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